

Ques:03 Prove that product of an even integer and an odd integer is even.

Suppose m is an even integer and n is an odd integer. Then,

$$m = 2k, \quad n = 2l + 1$$

Now

$$m \cdot n = (2k)(2l + 1) = 2 \cdot k(2l + 1)$$

$$= 2r$$

where $r = k(2l + 1)$ is an int.

Hence $m \cdot n$ is even. (proved)

Ques:04 Prove that square of an even int. is even.

Let $n = 2k$ be an even integer.

$$\text{Square of } n = n^2 = (2k)^2 = 4k^2 = 2(2k^2)$$

$$= 2p \quad \text{where } p = 2k^2 \in \mathbb{Z}$$

Hence, n^2 is even. (proved)

Ques: 05

Prove that if n is an odd integer, then $n^3 + n$ is even.

Let n be an odd integer. Then,
 $n = 2K + 1$ for some $K \in \mathbb{Z}$.

Now

$$\begin{aligned} n^3 + n &= n(n^2 + 1) \\ &= (2K + 1)[(2K + 1)^2 + 1] \\ &= (2K + 1)[4K^2 + 1 + 4K + 1] \\ &= (2K + 1)(4K^2 + 4K + 2) \\ &= (2K + 1) \cdot 2(2K^2 + 2K + 1) \\ &= 2 \cdot (2K + 1)(2K^2 + 2K + 1), \quad K \in \mathbb{Z} \\ &= \text{an even integer} \end{aligned}$$

Ques:11 Prove that sum of any three consecutive integers is divisible by 3.

Let $n, n+1, n+2$ be three consecutive integers.

Now

$$\begin{aligned}n + (n+1) + (n+2) &= 3n + 3 \\ &= 3(n+1)\end{aligned}$$

$$= 3 \cdot k \quad \text{where } k = n+1 \in \mathbb{Z}$$

Hence, the sum of three consecutive integers is divisible by 3.